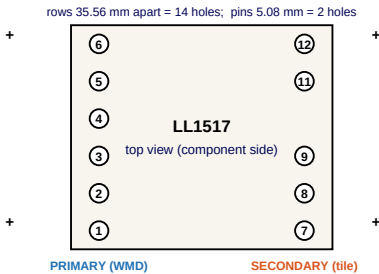


FOH isolation board: bench sheet, v3 (measured headers, no jumpers)

Two Lundahl LL1517 transformers on a 48 x 17 hole perfboard (122 x 43 mm), sitting in the ribbon between the WMD's rear balanced-out header and the Stereo Out Jacks 1U tile. 1:1, passive, galvanically isolated. Four signals cross through iron; no ground ever crosses. No insulated jumpers: every wire is bare and nothing crosses.

1. LL1517 pin key



Pin	What it is	T1 (LEFT), mirrored	T2 (RIGHT)
1	primary A start (+)	IN L- (cold)	IN R+ (hot)
3	primary A end	bridge to 6	bridge to 6
6	primary B start (+)	bridge from 3	bridge from 3
4	primary B end	IN L+ (hot)	IN R- (cold)
2, 5	centre taps	nothing	nothing
7	secondary A start (+)	OUT L- (ring)	OUT R+ (tip)
8	secondary A end	bridge to 12	bridge to 12
12	secondary B start (+)	bridge from 8	bridge from 8
11	secondary B end	OUT L+ (tip)	OUT R- (ring)
9	core + housing	OUT GND (45,9)	OUT GND (45,9)

T1 is wired mirrored on purpose: both primary and secondary are reversed. Reversing one side alone would flip that channel; reversing both cancels out, so L and R stay in phase. It is what lets the wires reach the headers without crossing. (In v2 it was T2; in v3 it is T1.) Bridges are the same on both: 3-6 and 8-12, never 3-4.

2. Header pin maps (measured 13 Sept)

Both headers measured the same, viewed from the component side: top row GND, GND; middle row L+, L-; bottom row R+, R-. On this board the IN header is fitted with its GND pins toward the top edge (row 7) and the OUT header with its GND pins toward the BOTTOM edge (row 9), i.e. turned 180°, so the holes read as below. Column 0 is the IN end, row 0 the top edge. IN header = columns 1-2, OUT header = columns 45-46, both in rows 7-9.

IN (WMD)	col 1	col 2	OUT (tile)	col 45	col 46
row 7	GND, no wire	GND, no wire	row 7	R-	R+
row 8	L+	L-	row 8	L-	L+
row 9	R+	R-	row 9	GND (both shields)	GND, no wire

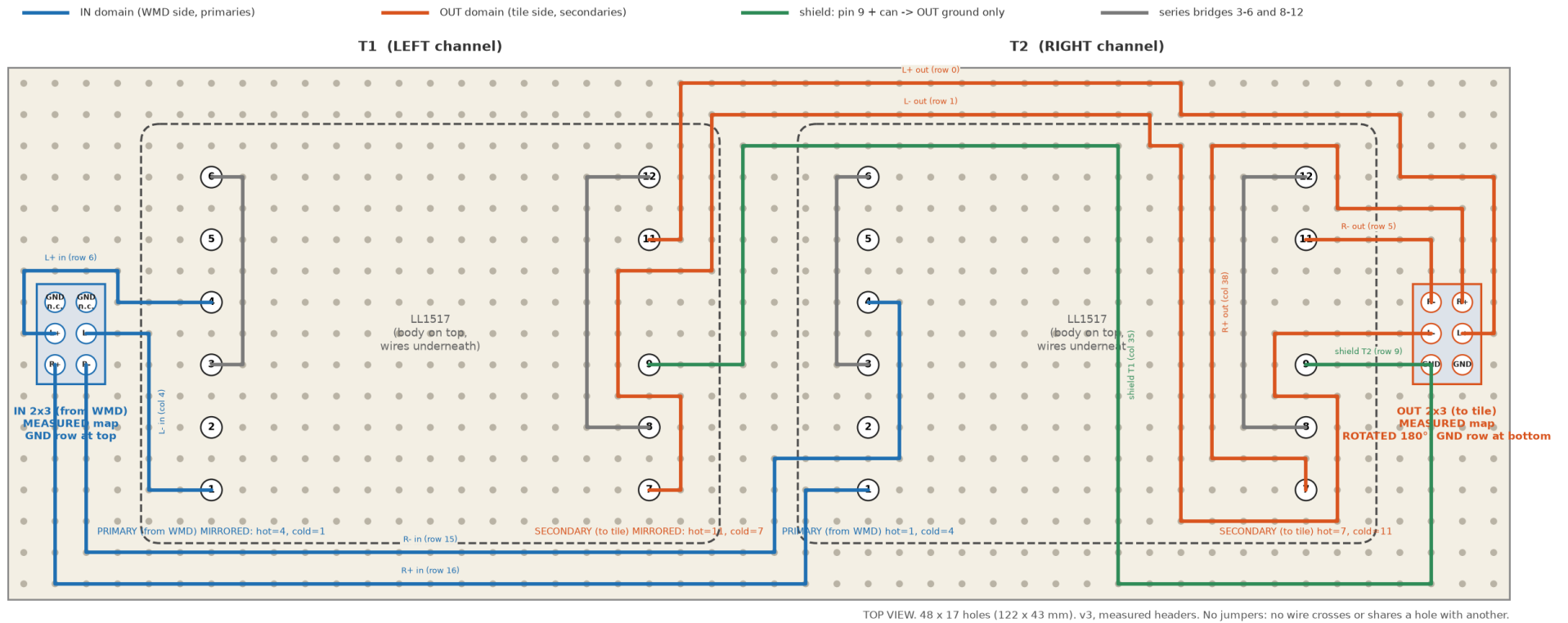
Orientation: fit the IN header with its two GND pins toward the top edge (row 7), and its notch facing the same way as the notch on the WMD does when you hold the WMD with its GND pins up. Then fit the OUT header TURNED 180°: its two GND pins toward the bottom edge (row 9), and its notch facing the same way as the notch on the tile does when you hold the tile with its GND pins DOWN. The OUT header is turned round compared with v2 and with the IN header; the ribbon plug turns with it, so every signal still lands where the drawing expects it. Step B4 checks this with the meter before any wiring. Both ribbons pin 1 to pin 1, stripe on the same side at both ends; make them different lengths.

3. Sharpie key for the underside

BLUE	WMD side: L+, L-, R+, R-. Long runs: L+ row 6, L- col 4, R+ row 16, R- row 15.
ORANGE / RED	Tile side: L+, L-, R+, R-. Long runs: L+ row 0, L- row 1, R+ col 38, R- row 5.
GREEN	Shields: pin 9 of each transformer to OUT GND (45,9). Longest runs: T1 col 35, T2 row 9.
GREY / BLACK	Bridges 3-6 and 8-12: T1 3-6 on col 7, T1 8-12 on col 18, T2 3-6 on col 26, T2 8-12 on col 39. No jumpers anywhere.

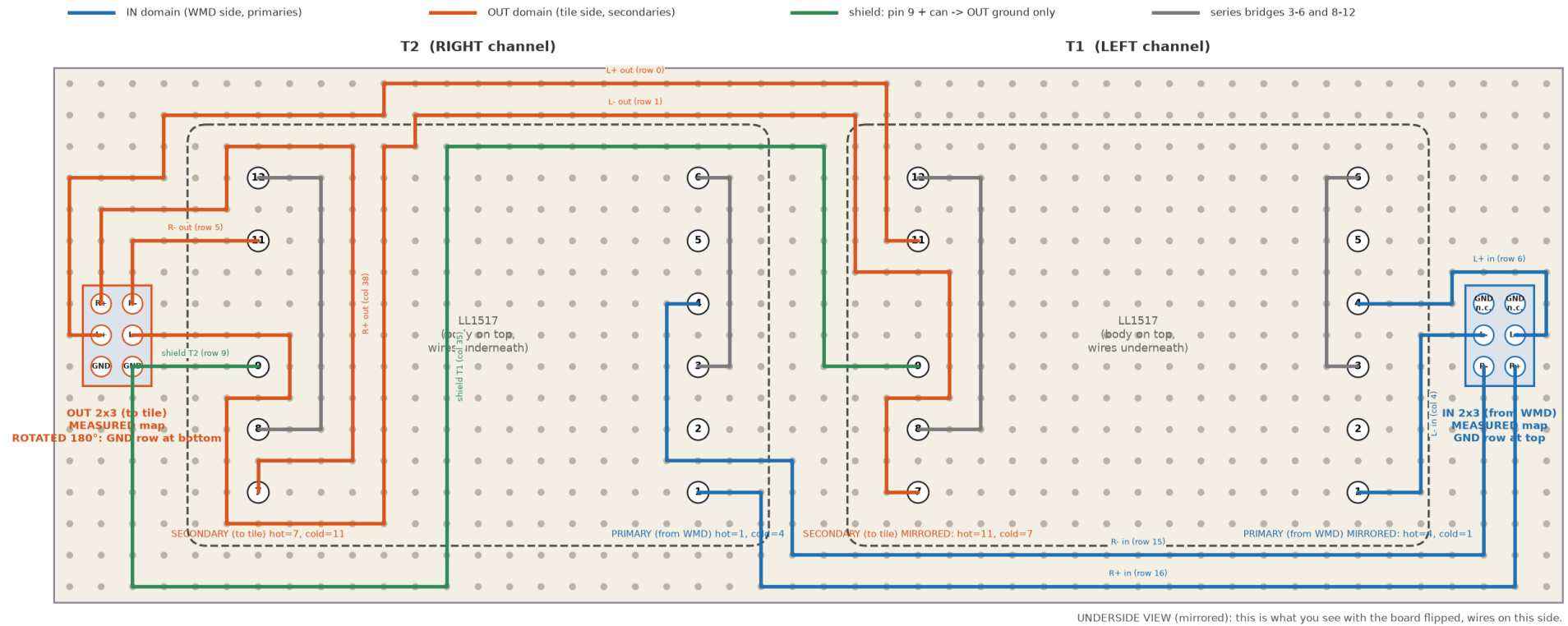
Draw the lines in the gaps beside the holes, not through the pad rings. Ink under a joint still solders; it just looks scruffy. A blue wire touching an orange one anywhere is a failed board.

Routing, TOP view (component side). Column 0 is the IN end, row 0 is the top edge.



v3 for the measured headers. Bodies on top, every wire underneath. Machine-checked (solver/check_routes.py): no wire crosses another and no two wires share a hole, except the two shield wires that end on the same GND pin.

Routing, UNDERSIDE view (mirrored). This is what you see with the board flipped, wires facing you.



Use this page at the iron. The IN end is now on the RIGHT. Lay silver wire flat from pad to pad along the drawn path; solder each pad it lands on; do not let bare wire cross another bare wire.

4. Build order

The directions give hole coordinates (col,row). Up and down mean toward row 0 and row 16; they are the same on both drawings.

A. Prepare

- [] Cut the board to 48 x 17 holes (122 x 43 mm). Score both faces along a hole row with a knife against a rule, snap over a table edge, tidy with P120 on the block. Backup: junior hacksaw (32 TPI). Wipe the glass dust off.
- [] Mark the two transformer footprints on TOP: primary column at col 6 (T1) and col 27 (T2), secondary column 14 holes on, at col 20 and col 41. Pins at rows 3, 5, 7, 9, 11, 13 (primary) and 3, 5, 9, 11, 13 (secondary). Check a transformer physically against the marks before drilling.
- [] Drill the 22 footprint holes to 1.5 mm with the HOTO at 600 rpm, light pressure. The existing hole centres the bit. All other holes stay 1.0 mm.
- [] Sharpie the routing on the UNDERSIDE using page 3 and the colour key.

B. Headers and ribbons

- [] Fit both 2x3 box headers from the top: IN at cols 1-2 rows 7-9, OUT at cols 45-46 rows 7-9, oriented as in section 2 (OUT header turned 180°, GND pins toward the bottom). Tack one pin, check it sits flat, solder the rest.
- [] Peel 6 conductors off the 10-way ribbon. Cut square. Crimp a PFL 6 on each end in the vice: stripe over pin 1, cap pressed home parallel with scrap card either side, strain relief folded over. Two short ribbons (different lengths) plus one longer bypass ribbon.
- [] Test each ribbon: all six conductors continuous end to end; no continuity between neighbouring pins.
- [] Orientation check, before any wiring: plug the tile's ribbon into the OUT header. Meter from a tile jack sleeve to the OUT header holes: expect 0 ohm at (45,9) and (46,9). If the 0 ohm readings are at (45,7) and (46,7) instead, the header is the wrong way round: stop and refit it.

C. Transformers and wiring

- [] Seat T1 and T2 from the top with the primary pins toward the IN end. Every pin drops in without force. Solder all 11 pins on each; short passes.
- [] Bridges first (grey): T1 3-6 on col 7, T1 8-12 on col 18, T2 3-6 on col 26, T2 8-12 on col 39.
- [] **T1 is mirrored: IN L+ goes to pin 4 and IN L- to pin 1; OUT L+ comes from pin 11 and OUT L- from pin 7.**
- [] **IN L+** (blue) from IN L+ (1,8): along row 8 to col 0, up col 0 to row 6, along row 6 to col 3, down col 3 to row 7, along row 7 into T1 pin 4 (6,7).
- [] **IN L-** (blue) from IN L- (2,8): along row 8 to col 4, down col 4 to row 13, along row 13 into T1 pin 1 (6,13).
- [] **IN R+** (blue) from IN R+ (1,9): down col 1 to row 16, along row 16 to col 25, up col 25 to row 13, along row 13 into T2 pin 1 (27,13).
- [] **IN R-** (blue) from IN R- (2,9): down col 2 to row 15, along row 15 to col 24, up col 24 to row 12, along row 12 to col 28, up col 28 to row 7, along row 7 into T2 pin 4 (27,7).
- [] **OUT L+** (orange) from T1 pin 11 (20,5): along row 5 to col 21, up col 21 to row 0, along row 0 to col 37, down col 37 to row 1, along row 1 to col 44, down col 44 to row 3, along row 3 to col 47, down col 47 to row 8, along row 8 into OUT L+ (46,8).
- [] **OUT L-** (orange) from T1 pin 7 (20,13): along row 13 to col 21, up col 21 to row 10, along row 10 to col 19, up col 19 to row 6, along row 6 to col 22, up col 22 to row 1, along row 1 to col 36, down col 36 to row 2, along row 2 to col 37, down col 37 to row 14, along row 14 to col 42, up col 42 to row 10, along row 10 to col 40, up col 40 to row 8, along row 8 into OUT L- (45,8).
- [] **OUT R+** (orange) from T2 pin 7 (41,13): up col 41 to row 12, along row 12 to col 38, up col 38 to row 2, along row 2 to col 42, down col 42 to row 4, along row 4 to col 46, down col 46 into OUT R+ (46,7).
- [] **OUT R-** (orange) from T2 pin 11 (41,5): along row 5 to col 45, down col 45 into OUT R- (45,7).
- [] **T1 shield** (green) from T1 pin 9 (20,9): along row 9 to col 23, up col 23 to row 2, along row 2 to col 35, down col 35 to row 16, along row 16 to col 45, up col 45 into OUT GND (45,9).
- [] **T2 shield** (green) from T2 pin 9 (41,9): along row 9 into OUT GND (45,9).
- [] Both shield wires end on (45,9): solder them together on that pin.
- [] No wire on: IN GND pins (1,7) and (2,7); OUT GND (46,9); pins 2 and 5 of each transformer.

D. Bench tests, before it goes anywhere near the case

Meter between	Expect	Means
T pin 1 and pin 4 (each transformer)	about 18 ohm	primaries in series, bridge good
T pin 7 and pin 11 (each transformer)	about 19 ohm	secondaries in series, bridge good
T pin 1 and pin 7	OPEN	no copper between the sides
T pin 9 and pin 1; pin 9 and pin 7	OPEN	housing touches no coil

IN header GND and OUT header GND	OPEN	the isolation itself
every IN signal pin and every OUT pin	OPEN	no stray bridge across domains
IN L+ and IN L-; IN R+ and IN R-	about 18 ohm	header to primary correct (T1 mirrored)
OUT L+ and OUT L-; OUT R+ and OUT R-	about 19 ohm	secondary to header correct
OUT GND (45,9) and T1 pin 9; and T2 pin 9	0 ohm	both shields landed
OUT GND and each OUT signal hole (45,7), (45,8), (46,7), (46,8)	OPEN	no shield wire touching a signal wire

E. Sleeve, mount, install

- [] Cut about 150 mm of the 70 mm ISOLATECH tube. Slide on with both ribbons plugged, shrink from the centre out with the gun moving. Ends stay open.
- [] Mark IN and OUT on the sleeve with a paint pen.
- [] Two EPDM runners (about 120 mm, 15 x 3 mm) stuck to the case where the board sits, about 30 mm apart. Two tie bases on isopropyl-wiped aluminium either side, pressed 30 s, left 24 h. Zip tie through each base; ONE-WRAP strap through the zip ties and over the board.
- [] Installed, strap tight, NO FOH cable: tile jack sleeve to any WMD jack sleeve = OPEN, and tile jack sleeve to bare case metal = OPEN. Press and wiggle the board while watching the meter.
- [] Ribbon in, tone playing: turn MSTR, level at the tile follows. Heaviest track, full MSTR, into a desk: bottom end clean, L and R equal and in phase.
- [] Straight bypass ribbon into the flight case. Label the tile ISO OUT L/R, LINE LEVEL.